

RATIONALISED PRE-TECHNICAL STUDIES NOTES GRADE 9



STRAND: FOUNDATION OF PRE-TECHNICAL STUDIES.

SUB-STRAND:SAFETY ON RAISED PLATFORMS.

- A raised platform in the workplace is any elevated surface or structure where workers stand to perform tasks.
- It could be something like a ladder, scaffold, or lift that helps them reach higher areas safely.
- Examples or Types of Raised platforms; ladders, trestles steps,stands mobile raised platforms,work benches and ramps.

Roles of the ladder,trestles,steps,ramps,stands,work benches as raised platforms

1.Ladder:

- ★ Provides a portable way for workers to reach elevated areas, such as shelves or high walls.

2.Trestles:

- ★ Used to support work surfaces or planks for tasks like painting or construction, giving workers a stable

elevated surface.

3.Steps:

- ★ Offers a small, portable platform for reaching slightly elevated areas, useful for quick, short tasks.

4.Ramps:

- ★ Allows easy access for people or equipment to move between different levels, often used for loading and unloading.

5.Stands:

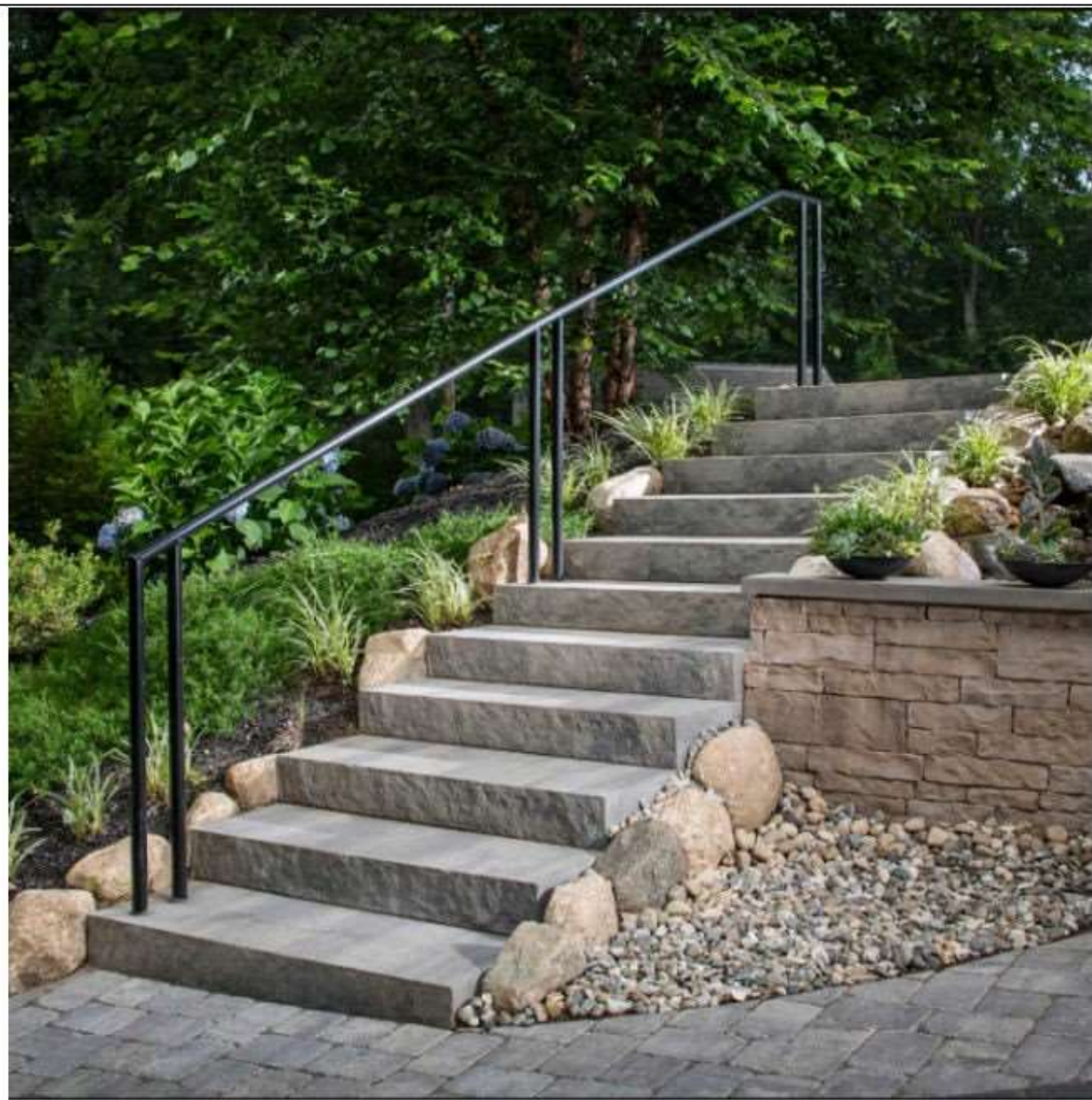
- ★ Provides a stable, elevated support for tools, equipment, or materials, often adjustable to different heights.

6.Work Benches:

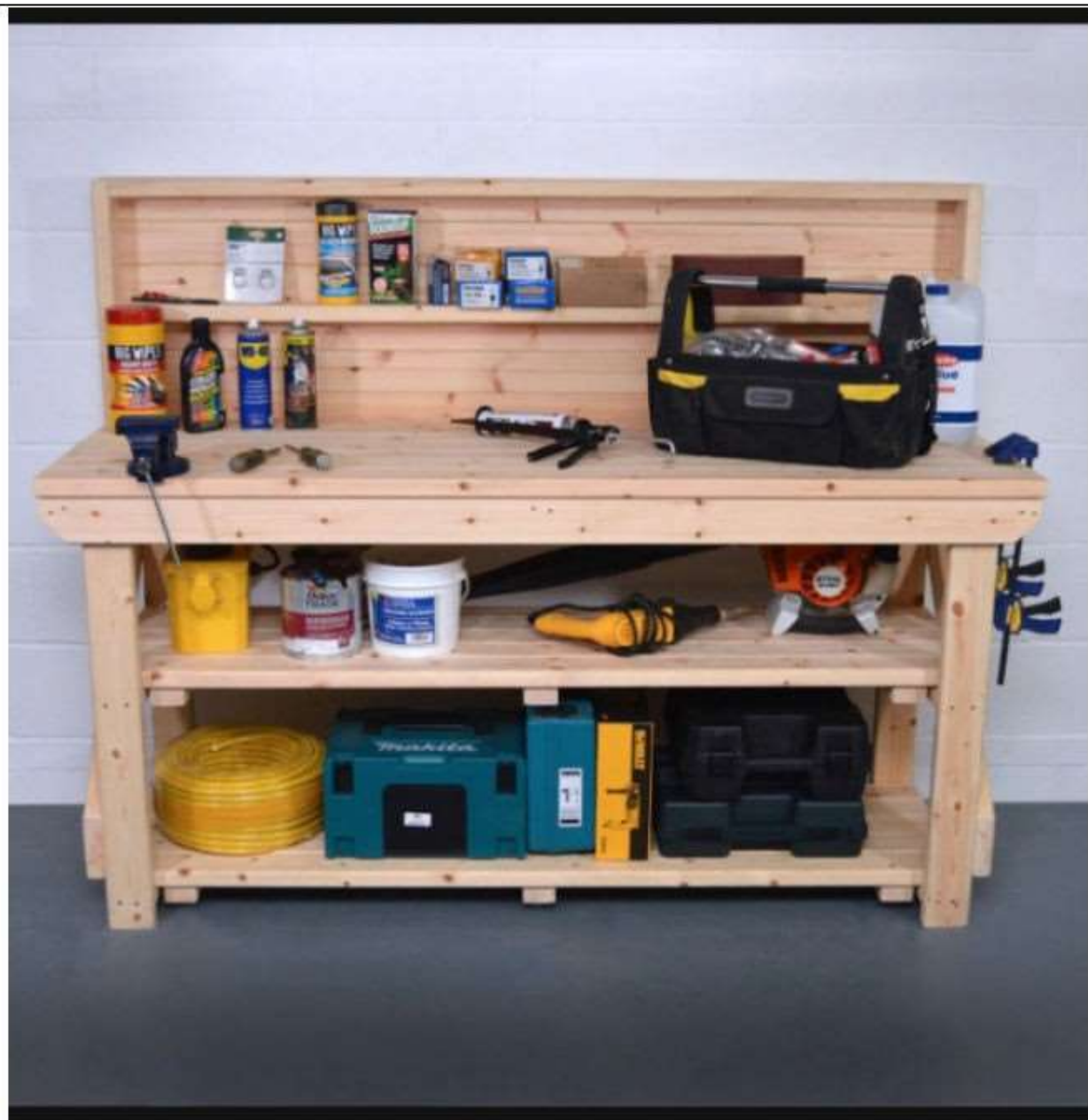
- ★ Serves as an elevated surface for performing tasks like assembling, cutting, or working with materials at a comfortable height.

PICTURES.













Risks associated with working on raised platforms.

Definition of Risk:

- Risk is the chance or possibility that something bad or harmful could happen. It's the likelihood of facing danger, injury, or loss.

Risks Associated with Working on Raised Platforms:

1. Falling from Heights:

- ★ The biggest risk is falling off the platform. This can

lead to serious injuries or even death.

2. Slips and Trips:

★ Raised platforms can become slippery due to rain, oil, or loose objects, increasing the risk of losing balance.

3. Equipment or Tools Falling:

★ Tools or materials can fall off the platform, possibly injuring people below.

4. Platform Collapse:

★ If the platform is overloaded or not set up properly, it can collapse, putting everyone on it in danger.

5. Electrocution:

★ If the platform is near power lines, there is a risk of electric shock or electrocution.

6. Weather-Related Risks:

★ Strong winds, rain, or ice can make working on raised platforms more dangerous by affecting balance and stability.

7. Unstable Surface:

★ If the ground or surface where the platform is set up is uneven or soft, it can shift or tip, increasing the chance of accidents.

8. Lack of Proper Guardrails:

★ Without guardrails or safety barriers, there's a higher chance of falling off the edges of the platform.

9. Limited Visibility:

★ Poor lighting, fog, or dust can make it harder to see, increasing the chances of missteps or slips.

10.Fatigue:

- ★ Working at heights can be physically and mentally tiring, which might lead to mistakes or slower reactions to hazards.

11.Improper Use of Safety Gear:

- ★ If workers don't wear harnesses correctly or use the wrong equipment, the effectiveness of safety measures is reduced.

12.Overcrowding:

- ★ Too many people on a small platform can make it unstable, increasing the risk of falls or tipping.
- Taking steps to understand and control each of these risks can greatly improve safety when working on raised platforms.

Ways of minimizing risks related to working on raised platforms.

Here are some ways to minimize these risks:

1.Use Safety Gear:

- ✓ Always wear safety harnesses and helmets to protect yourself in case of a fall.

2.Check Equipment:

- ✓ Ensure that the platform, ladder, or scaffold is stable, secure, and in good condition before use.

3.Keep the Area Clean:

- ✓ Remove any objects that could cause someone to slip or trip. Keep the platform dry to prevent slips.

4. Don't Overload the Platform:

- ✓ Make sure the platform can safely hold the weight of people and equipment on it.

5. Use Guardrails:

- ✓ Install guardrails or safety barriers on raised platforms to prevent falls.

6. Training:

- ✓ Make sure everyone using the platform has proper training on how to use it safely.

7. Watch Out for Weather Conditions:

- ✓ Avoid working on raised platforms during strong winds or rain, as this can make the platform more dangerous.

- By following these precautions, the risk of accidents and injuries can be greatly reduced.

Importance of observing safety when working on raised platforms.

1. Prevents falls and serious injuries.
2. Ensures a stable working environment.
3. Reduces the risk of equipment or tools falling.
4. Promotes compliance with safety regulations.
5. Protects workers and others around the platform.

- 6.Increases overall productivity by preventing accidents.
- 7.Reduces the likelihood of costly medical or legal issues.
- 8.Enhances worker confidence when safety is prioritized.
- 9.Protects against structural collapse of the platform.
- 10.Ensures proper use of safety gear, like harnesses and guardrails.

SUB-STRAND 2.

HANDLING HAZARDOUS SUBSTANCES.

- Hazardous Substances: These are materials that can cause harm to people, animals, or the environment if not handled carefully. They may be poisonous, flammable, or corrosive.
- Poisonous: Substances that can make you sick or even be deadly if you eat, drink, breathe in, or touch them. Examples include pesticides or certain chemicals used in cleaning.



- **Flammable:** These are substances that can easily catch fire if exposed to heat or a spark. Examples include gasoline, alcohol, and paint thinners.



- **Corrosive:** These are substances that can cause damage by wearing away or "eating through" materials, including skin or metal. Examples include strong acids like battery acid and bleach.



- Household products containing strong acids or bases.

Safe Ways to Handle Hazardous Substances

1. **Wear Protective Gear:** Use gloves, masks, goggles, or even aprons when handling hazardous substances to avoid skin contact or inhalation.
2. **Proper Storage:** Keep hazardous materials in clearly labeled, sealed containers and store them in a cool, dry place, away from sunlight and sources of heat.
3. **Avoid Mixing Chemicals:** Some substances, like bleach and ammonia, create toxic fumes when mixed, so it's essential to keep them separate.
4. **Follow Disposal Instructions:** Dispose of hazardous waste according to local guidelines to prevent contamination of soil or water. Never pour chemicals down the drain unless it's safe to do so.
5. **Ventilate When Using Indoors:** Open windows or doors to allow fresh air to circulate, especially when using flammable or strong-smelling products.
6. **Use Only in Well-Ventilated Areas:** When handling substances that release fumes, work outdoors if possible or in areas with good airflow.

Importance of Observing Safety When Handling Hazardous Substances

1. **Protects Health:**

- Proper handling prevents exposure to toxins that could cause serious health issues like poisoning, respiratory problems, burns, or even cancer.

2.Environmental Protection:

- Correctly storing and disposing of hazardous substances prevents pollution of air, soil, and water, protecting wildlife and plant life.

3.Prevents Accidents and Fires:

- Following safety measures reduces the risk of accidental fires or chemical reactions that could harm people and property.

4.Legal Compliance:

- Observing safety rules helps individuals and companies follow laws designed to protect people and the environment from harm.

5.Promotes Community Safety:

- When everyone follows safety guidelines, it ensures that hazardous substances do not pose risks to the larger community.

SELF-EXPLORATION AND CAREER DEVELOPMENT.

Self-Exploration:

- This means discovering your own interests, strengths, and what you enjoy doing. It's learning about yourself to make better choices for your future.

Self-exploration is necessary for career development because;

- it helps you understand who you are, what you enjoy, and what you're good at.
- This knowledge is important because it guides you in choosing a career that makes you happy and successful.

When you explore yourself, you can:

- Identify Strengths and Interests: Knowing what you're good at and what excites you helps you focus on careers that match your abilities.
- Set Realistic Goals: Understanding yourself makes it easier to set achievable career goals that align with your skills and passions.
- Increase Confidence: When you know your talents,

you feel more confident in your abilities and decisions.

- **Make Better Choices:** Self-exploration helps you avoid careers that might make you unhappy or stressed. Instead, you can choose paths that motivate and inspire you.
- In short, self-exploration is like a map that helps you find the right career journey for yourself.

Career Development:

- This is the process of learning and planning for a job or career that fits your skills and interests.
- It includes studying, getting experience, and setting career goals.

Talents and Abilities.

- A Talent is a natural ability that allows you to do a particular task with or without guidance.
- An Ability is a skill that a person learns through a lot of practice.

Differences between Talent and Ability.

TALENT	ABILITY/SKILL
Gift of God.	Acquired and developed
Possessed by few people.	Many people can possess it through learning.
Requires recognition.	Requires development.

Requires guidance and coaching.	Requires training.
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Ways of Nurturing Talents and Abilities

1. Practice Regularly: Spend time improving your skills through practice. For example, if you're good at art, draw regularly.
2. Seek Guidance: Find a mentor, coach, or teacher who can help you improve. They can give you tips and help you grow your talent.
3. Set Goals: Set small goals to reach a bigger goal. This helps you stay focused and keep improving.
4. Get Training or Education: Take courses, workshops, or extra classes to learn more about your area of interest.
5. Find a Supportive Community: Join groups or clubs where people have similar interests. This helps you learn from others and stay motivated.
6. Observing and identifying a talent.
7. Providing resources to cultivate interests on talents and abilities.

Relating Talents and Abilities to Career Pathways.

1. STEM (Science, Technology, Engineering, Mathematics):

- If you are good with numbers, problem-solving, or technology, STEM careers could be a fit. Talents in logical thinking, designing, or building can lead to careers in engineering, medicine, IT, or research.

2.Social Science:

- If you enjoy helping people, solving social issues, or studying human behavior, social sciences might be for you.
- Talents like empathy, communication, and critical thinking are useful in careers like teaching, psychology, law, and social work.

3.Creative Arts:

- If you love drawing, acting, singing, or creating things, creative arts could be a good pathway. Talents in artistic expression, imagination, and storytelling can lead to careers in music, art, drama, and media.

Career Opportunities Related to Talents and Abilities

- 1.Musical Talent: Careers like musician, music teacher, sound engineer.
- 2.Artistic Talent: Careers like artist, graphic designer, animator, or interior designer.
- 3.Mathematical Ability: Careers like accountant, data analyst, engineer.

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